

Memory-Centric Agentic Inference — cycles 4-6 clone 0

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Abstract

Cycles 4-6 converted the already validated DC-005 trajectory reuse measurement branch from a completed design into a conductor-ready merge package. No new trajectory reuse semantics were introduced. The main work was to make merge acceptance observable: a deterministic verifier now checks that the parent measurement CSVs still contain the required TRJ-* experiment rows, collapse thresholds, required production fields, and the boundary that keeps DC-006 provenance-overhead measurement out of DC-005.

Cycle 4 built and validated the merge-readiness package. Cycle 5 defined the post-merge acceptance, ledger-repair, and reopen conditions. Cycle 6 closed the clone at research scope with a terminal handoff. The final audit decision for cycles 4-6 was **VALIDATED**: the clone has a validated prose plan, validated parent harness rows, a validated verifier, and clear conductor instructions.

The only unresolved item is process-level, not semantic: before conductor reconciliation, root `promise_check` reports orphan artifacts because fanout clones write through shadow ledgers. Auditors treated those warnings as expected pre-merge ledger isolation while the verifier and organization checks pass.

Introduction

The root project investigates whether agentic inference infrastructure should be organized around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. In that program, three architecture options recur:

- **Option A**: conventional request/model/KV-centric serving.
- **Option B**: a memory-object-aware runtime.
- **Option C**: a trajectory/DAG-aware memory fabric, where a DAG is a directed acyclic graph representing branch, merge, replay, verification, and durable workspace relation-

ships across an agent run.

DC-005 is the deferred constant for production trajectory reuse distributions. It asks whether real agent traces contain enough reusable branch state, verifier state, tool-output replay, trajectory-log replay, and durable workspace dependency reuse to justify Option C over Option B or A.

Cycles 1-3 had already produced and validated the required artifact, `memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md`, and integrated it into parent M-EXP-1 CSV harness files. Cycles 4-6 therefore focused on preserving that validated content through conductor merge, not on redesigning the measurement branch.

Methodology

The cycles 4-6 methodology was conservative. Researchers treated the DC-005 plan and CSV rows as fixtures. Workers were directed not to reinterpret TRJ-* semantics and not to absorb sibling deferred constants into this branch. Auditors then checked whether each cycle maintained that boundary.

The work followed a three-step sequence:

1. Cycle 4 built a merge-readiness verifier.
2. Cycle 5 converted verifier results into conductor acceptance and reopen criteria.
3. Cycle 6 issued a terminal handoff and confirmed no clone-local work remained.

The key distinction introduced in these cycles is between two failure classes:

- **Content/schema failure:** missing artifacts, corrupt CSVs, missing TRJ-* rows, missing thresholds, missing required fields, or DC-006 leakage into DC-005. These require reopening DC-005.
- **Ledger-registration warning:** `root promise_check` reports orphan artifacts before fanout shadow-ledger reconciliation. This is a conductor ledger issue if the verifier and organization checks pass.

Results

Cycle 4: Merge-Readiness Verifier

Cycle 4 researcher session `d1be6a2b-779c-427d-afcf-2da43e00de5d` pivoted from trajectory reuse design to conductor merge verification. The research brief stated that the validated DC-005 plan, parent CSVs, and merge report should remain frozen. The useful remaining work was a verification package that could distinguish expected fanout ledger warnings from real artifact loss.

Worker session `6eb105b2-33a0-4228-af71-e1194bf964c5` built three artifacts:

Artifact	Role
<code>tests/verify_dc005_merge_ready.py</code>	Deterministic verifier for DC-005 merge readiness.
<code>memory-centric-agentic/experiments/dc005_merge_verification.md</code>	Human-readable conductor instructions, expected warnings, and reopen conditions.
<code>data/dc005_merge_verification_results.csv</code>	PASS/FAIL result rows emitted by the verifier.

The verifier reads five parent harness CSVs:

- data/measurement_experiment_specs.csv
- data/measurement_required_fields.csv
- data/measurement_thresholds.csv
- data/measurement_claim_update_matrix.csv
- data/measurement_synthetic_probe_results.csv

It then checks that the CSVs exist, parse, and contain at least one data row. It also checks for:

- all required experiment IDs, TRJ-001 through TRJ-007;
- all required thresholds: C_to_B_trajectory_reuse, C_to_A_trajectory_reuse, p_survive_min, and p_verifier_reuse_min;
- required fields: trajectory_node_id, replay_authorization_scope, verifier_evidence_hash, and retention_hold_state;
- no deferred_constant=DC-006 rows in the DC-005 merge package;
- at least one scope-boundary row confirming authorization/provenance fields are validity gates, not overhead measurements.

Worker validation reported that the verifier passed 10 checks with 0 failures, compiled cleanly, and detected an isolated negative probe where retention_hold_state was removed.

Auditor session 020362da-17b3-457a-88c1-167001d9c997 validated the cycle with no critical or moderate issues. The only minor note was expected pre-merge orphan warnings from root promise_check.

Cycle 5: Post-Merge Acceptance Criteria

Cycle 5 researcher session 67ec63c1-c1cb-47a1-a668-2dd5e38a625b moved DC-005 from clone-local work to conductor post-merge acceptance. The brief recommended no new build. Instead, it defined the conductor's acceptance commands:

```
python3 tests/verify_dc005_merge_ready.py
python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace
python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace
```

Worker session 97d29180-3d19-42c5-8b04-089eb200b9b8 built no new artifacts. It restated the binary conductor gate:

Outcome	Condition
Accept DC-005	Verifier passes, organization check is green, and root ledger recognizes DC-005 artifacts after reconciliation.
Route to ledger repair	Verifier and organization check pass, but artifact-registration warnings persist.
Reopen DC-005	Required plan, rows, thresholds, fields, or CSVs are missing/corrupt, or DC-006 leaks into the DC-005 measurement package.

Auditor session 8d3abccd-7b4d-4338-b5af-c84e86823211 reran the key checks. The verifier passed 10/10 checks. org_check passed. promise_check produced warnings only, matching the known pre-merge orphan artifact state. The audit decision was VALIDATED.

Cycle 6: Terminal Handoff

Cycle 6 researcher session db297ecb-f02e-4dbd-86ff-318242a6659e declared clone 0 complete at research scope. It stated that the prose plan, parent harness CSV integration, merge-readiness verifier, and post-merge acceptance criteria were all validated.

Worker session c14b7cff-400b-439d-99fb-cf9c81a69588 built no new artifacts and reran no commands. The worker confirmed that the conductor should preserve the validated artifacts and perform merge reconciliation.

Auditor session 213b824f-0207-49b7-9c1b-abab007e71e9 validated the terminal handoff. The supplied audit report recorded:

- `python3 tests/verify_dc005_merge_ready.py`: PASS, 10/10 checks.
- `python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace`: PASS.
- `python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace`: warnings only, consistent with known pre-merge orphan state.

The audit found no critical or moderate defects. It classified the remaining warnings as ledger-registration warnings expected before conductor reconciliation. The final decision was VALIDATED.

Findings

The main result of cycles 4-6 is that DC-005 is no longer just a validated measurement design. It is also a merge-verifiable package. The conductor can now test whether the validated design survived merge by running one deterministic script and two workspace checks.

The verifier makes the branch's scope boundary executable. DC-005 may include authorization and provenance fields as replay-validity gates. It may not measure DC-006 provenance-validation overhead magnitude. A DC-006 measurement row inside the DC-005 CSV package is a verifier failure.

The reopen conditions are concrete. DC-005 should reopen only for missing or corrupt artifacts, missing TRJ-* rows, missing thresholds, missing required fields, or cross-constant contamination. It should not reopen for known pre-merge fanout ledger warnings while the verifier and organization checks pass.

The cycles also clarified the terminal state of the clone. Further clone-local work would be a null cycle. The next useful action belongs to the conductor: merge reconciliation, verifier rerun, organization check, and root promise ledger check.

Discussion

Cycles 4-6 changed the risk profile from research uncertainty to integration observability. Before these cycles, the branch had validated content but still depended on human inspection to decide whether merge warnings were harmless. After cycle 4, the branch had a deterministic verifier. After cycles 5 and 6, the conductor had a decision procedure for acceptance, ledger repair, or reopen.

This matters for the memory-centric architecture program because DC-005 is a gating constant for Option C. If trajectory reuse rows are lost, thresholds are corrupted, or provenance-overhead work leaks into this branch, downstream architecture comparisons can claim trajectory/DAG benefits incorrectly. The verifier prevents that class of silent degradation.

The remaining warning is intentionally not treated as a DC-005 defect. Fanout clones use shadow ledgers, so `root promise_check` can report clone-produced files as orphan artifacts be-

fore conductor reconciliation. The reportable decision from auditors is that these warnings are tolerated before merge only when the verifier passes and `org_check` is green.

Conclusions and Recommendations

DC-005 clone 0 is complete for conductor merge handling. Across cycles 4-6, the branch gained:

- a deterministic merge-readiness verifier;
- a verifier result CSV;
- conductor-facing documentation;
- validated post-merge accept, ledger-repair, and reopen conditions;
- terminal audit validation.

The conductor should accept DC-005 after merge if:

```
python3 tests/verify_dc005_merge_ready.py
python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace
python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace
```

all produce the expected accepted state: verifier pass, organization pass, and reconciled ledger registration.

If the verifier and organization check pass but only orphan warnings persist, the next action is conductor ledger repair. If the verifier fails, DC-005 should reopen only around the named missing artifact, row, threshold, field, corrupt CSV, or cross-constant contamination.

References

No external references were newly cited by this cycles 4-6 clone report. The report source material is the supplied session record, local DC-005 artifacts, and the supplied audit report.

Appendix: Implementation Details

Source Inventory

Source	Date	Contents	Timeline role
d1be6a2b-779c-427d-afcf-2da43e00de5d	2026-05-11	Research brief pivoting from closed DC-005 design to merge-readiness verification.	Cycle 4 research direction.
6eb105b2-33a0-4228-af71-e1194bf964c5	2026-05-11	Worker output building verifier, documentation, and result CSV.	Cycle 4 implementation.
020362da-17b3-457a-88c1-167001d9c997	2026-05-11	Audit validating verifier, negative probe, and expected ledger caveat.	Cycle 4 validation.

Source	Date	Contents	Timeline role
67ec63c1-c1cb-47a1-a668-2dd5e38a625b	2026-05-11	Research brief defining conductor post-merge acceptance criteria.	Cycle 5 research direction.
97d29180-3d19-42c5-8b04-089eb200b9b8	2026-05-11	Worker no-build handoff with accept, ledger-repair, and reopen gates.	Cycle 5 handoff.
8d3abccd-7b4d-4338-b5af-c84e86823211	2026-05-11	Audit validating the executable handoff gate.	Cycle 5 validation.
db297ecb-f02e-4dbd-86ff-318242a6659e	2026-05-11	Research terminal handoff for clone 0.	Cycle 6 research closure.
c14b7cff-400b-439d-99fb-cf9c81a69588	2026-05-11	Worker no-build terminal handoff.	Cycle 6 closure confirmation.
213b824f-0207-49b7-9c1b-abab007e71e9	2026-05-11	Final audit report validating terminal handoff.	Cycle 6 final decision.

Artifact Inventory

File	Rows or lines	Role
memory-centric-agentic/experiments/trajectory_reuse_measurement_plan.md	298 lines	Required DC-005 measurement-plan artifact inherited from cycles 1-3.
data/measurement_experiment_specs.csv	7 data rows	Parent harness experiment specs, TRJ-001 through TRJ-007.
data/measurement_required_fields.csv	13 data rows	Required trajectory, replay, authorization, verifier, and retention fields.
data/measurement_thresholds.csv	4 data rows	Option C collapse and minimum-reuse thresholds.
data/measurement_claim_update_matrix.csv	11 data rows	Claim update routing for DC-005 measurement outcomes.
data/measurement_synthetic_probe_results.csv	6 data rows	Synthetic mechanism probes, not production calibration.
tests/verify_dc005_merge_ready.py	208 lines	Deterministic DC-005 merge-readiness verifier added in cycle 4.

File	Rows or lines	Role
memory-centric-agentive/experiments/dc005_merge_verification.md	54 lines	Conductor-facing verification documentation added in cycle 4.
data/dc005_merge_verification_results.csv	10 data rows	Latest verifier result table.
MANIFEST.md	updated during reporting	Workspace snapshot refreshed to include the DC-005 verifier package.

Validation Summary

The audits for cycles 4-6 reported no critical or moderate defects.

Cycle 4 validation confirmed that the verifier exists, compiles, passes, writes 10 PASS rows, detects a missing required field in an isolated negative probe, and documents pre-merge ledger warnings.

Cycle 5 validation confirmed that the conductor can make a binary post-merge decision using the verifier, organization check, and promise ledger check.

Cycle 6 validation confirmed that the terminal handoff is sound. The verifier passed 10/10 checks, org_check passed, and promise_check produced only known pre-merge orphan warnings.

Cross-Reference Map

Origin	Consuming artifact	Flow
trajectory_reuse_measurement_plan.md	data/measurement_experiment_specs.csv	Prose measurement design becomes TRJ-001 through TRJ-007.
trajectory_reuse_measurement_plan.md	data/measurement_required_fields.csv	Required production fields become harness rows.
trajectory_reuse_measurement_plan.md	data/measurement_thresholds.csv	Option C collapse and minimum-reuse equations become harness thresholds.
data/measurement_*.csv	tests/verify_dc005_merge_ready.py	Verifier checks parseability, required rows, required fields, thresholds, and scope boundary.
tests/verify_dc005_merge_ready.py	data/dc005_merge_verification_results.csv	Verifier emits one row per check.
tests/verify_dc005_merge_ready.py	memory-centric-agentive/experiments/dc005_merge_verification.md	Documentation explains how conductor should run and interpret the verifier.
Cycle 4 audit	Cycle 5 handoff	Validated verifier becomes conductor acceptance mechanism.

Origin	Consuming artifact	Flow
Cycle 5 audit	Cycle 6 terminal handoff	Validated handoff gate becomes final clone closure basis.